Text-Augmented Reduced-Form and Structural Estimation of Corporate Default

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April 23, 2026

Abstract

With the rise of Natural-Language Processing (NLP), there has been a growing body of literature attempting to identify textual information in financial filings as previously latent variables that affect financial and corporate decision-making. Using a pre-trained neural language model (FinBERT) and Loughran-McDonald dictionary-based sentiment scores, I perform sentiment analysis on the Risk Factors and Management Discussion and Analysis (MD&A) sections of firm's SEC 10-K financial reports to study whether public financial disclosures carry latent information on firm quality and credit risk. Having collected text signals over a panel covering 63,449 firm-years on US public firms between 2006–2022 (including 125 default events), I find that the FinBERT signal is strongly significant in a linear probability model with financial controls and industry/year fixed effects, while LM signals are not. The top FinBERT quintile concentrates 86% of all subsequent default events in our sample. The result survives an out-of-sample test, is robust to inclusion of a Merton-naive distance-to-default control, and is not closed by constructing a supervised LM signal that mimics the FinBERT projection procedure. Along with this reduced-form analysis, I then replicated a version of a corporate default model developed by Hennessey and Whited (2007) and updated it with a text-augmenting element, in which competitive lenders update beliefs about next-period productivity using a public text signal to set the price of corporate debt. We solve the model using a deep equilibrium network (DEQN) and attempt structural estimation by simulated method of moments with 22 standard moments. Using three diagnostic procedures (multi-start SMM, line search at two simulation scales), I find that the text-informativeness is not identified from this moment specification, indicating a need for moments designed to isolate the text channel in future work.

1 Introduction

The text-and-finance literature, beginning with Loughran and McDonald (2011), documents that the language of corporate disclosures contains information about firm prospects that is not subsumed by accounting variables. Using new advancements in natural language processing (NLP), many authors subsequently analyzed how disclosure sentiments can predict various firm financial characteristics. One such example is Buehlmaier & Whited (2017), who use textual analysis on publicly disclosed financing difficulties in SEC 10K filings as a measurement for the financial constraints of firms, showing that textual information strongly captures firm characteristics associated with financial constraints, with this measure of constraint associated to higher stock returns. One area where textual/sentiment analysis has yet entered is in the realm of debt or corporate default. Publicly disclosed information in a firm's annual reports may signal information on risk or financing issues not covered by the financial data alone. Furthermore, certain text or language in these

documents, when observed by competitive lenders, may affect their decisions to lend money to the firm, creating for higher debt cost. Modern corporate default literature has been guided by Hennessey & Whited (2007) (Which we'll refer to as HW07), who models the firm's capital structure as the joint outcome of investment opportunities, productivity shocks, and external-financing frictions. Default emerges endogenously when realized net worth falls below an optimally-chosen threshold. In this paper, I use textual analysis of the Item 1A (Risk Factors) and Item 7 (MD&A) of 10K annual reports to construct text signals to study whether these signals predict corporate default. I do so first through a reduced-form analysis, where text signals are added to a linear probability model with data corporate default events. Secondly, I then attempt to replicate the structural model of corporate default produced in HW07. In this model, competitive lenders who price firm debt only see the firms current productivity z_t, and their net worth $\tilde{w}_t$. I augment this model to include a text signal s, constructed from the sentiment analysis, that allows lenders to see the future productivity state z_{t+1}. Given the expanded state space of this model, I employ a deep equilibrium network (DEQN) approach following Maliar et al. (2021); Azinovic et al. (2022) to solve this augmented-text model.

On the empirical side, I find that sentiment extracted from 10-K filings using FinBERT (Araci, 2019) —a pre-trained transformer model fine-tuned on financial text— predicts subsequent default in a panel of US public firms over 2006–2022. I find that a within-year standardized FinBERT signal corresponds to a 0.37 percentage point increase in 2-year default probability per standard deviation, against an unconditional 2-year default rate of 0.24% in our sample, with this relationship being statistically significant. The corresponding LM signal coefficient is not statically different from zero, indicating that the predictive content of 10-K language for default is captured by neural language representations but missed by dictionary methods.

On the theoretical side, I find that λ_{text}, which is the parameter on "text-informativeness" to future productivity, is not identified from our simulated method of moments specification at feasible simulation sizes. To test this, I perform three diagnostic procedures: a full Nelder-Mead optimization, multi-start SMM with varied initial values for λ_{text}, and one-dimensional line searches at simulation sizes that differ by a factor of eight. I find that the text-only loss component is essentially flat across $\lambda_{\text{text}} \in [0, 1]$, and this flatness is deterministic rather than a Monte Carlo artifact (correlation between baseline and large-sim loss vectors exceeds 0.99). This finding potentially suggests that the standard HW07 moment set is not sharp enough to identify text-informativeness parameters, though I acknowledge numerous limitations in the research, including restricted data types and model solver difficulties that understated important variables like the corporate default rate.

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2 Related Literature

This paper draws on three literatures. The first is dynamic corporate finance with external financing frictions. Hennessey & Whited (2007) estimate a structural model in which firms face costly equity

issuance, asymmetric corporate taxes, and bankruptcy costs, identifying the costs of external finance from observed firm decisions. Bazresch et al. (2018) extends this by providing a comprehensive treatment of the estimation methodology, including practical guidance on moment selection and identification. Our model preserves the HW07 structure on the firm side and augments it on the lender side via an information-economics device.

The second literature concerns textual analysis of corporate disclosures. Loughran and McDonald (2011) introduce the dictionary that bears their name and document that words classified as "negative" in a finance context predict negative returns and abnormal events. The literature has since grown substantially; we focus on the comparison between dictionary-based and neural representations because that comparison is informative about what language carries default-relevant signal. The neural-language literature in finance is younger. Araci (2019) introduces FinBERT, a domain-adapted BERT model. Subsequent work has applied transformer-based models to a variety of prediction tasks; our contribution within this literature is to provide a clean apples-to-apples comparison of dictionary and neural approaches for default prediction.

The third literature is the deep-learning approach to dynamic economic models, which we use to solve our . Maliar et al. (2021) provide a general framework for solving dynamic equilibrium models with neural networks. Azinovic et al. (2022) extend the approach to models with occasionally-binding constraints. Our DEQN solver follows their approach with adaptations specific to the HW07 structure (separate networks for value, policy, and the default threshold; smooth approximations of the default indicator to enable gradient-based training).

3 Data

3.1 Sample

The estimation sample consists of US public firms covered by both Compustat (financial statements) and CRSP (stock returns and delisting records) over the period 2006–2022. The starting year is determined by the SEC's December 2005 mandate of Item 1A (Risk Factors) in 10-K filings; the ending year is determined by the LoPucki Bankruptcy Research Database, whose coverage ends January 2023. We exclude financial firms (SIC 6000–6999) and utilities (SIC 4900–4999), as is standard in the corporate finance literature due to different mechanics and regulated capital structures which distort their results. The final panel contains 63,449 firm-years across 8,590 unique firms. On table 1, I report the sample composition alongside figure 1, which shows the distribution of the sample over time.

3.2 Default measurement

In this paper, I define a default event as a firm-year in which the firm either: (i) appears in the LoPucki Bankruptcy Research Database, or (ii) is delisted by CRSP with a delisting code in the range 400–599 (corresponding to involuntary delistings due to financial distress and liquidations). Of 1,565 distress-related delistings in CRSP over our sample period, 81 match firm-years in our

Table 1: Sample summary statistics.

Panel A: Sample composition

Sample period	2006–2022
Unique firms	8,590
Firm-year observations	63,449
Default events (broad sample)	125
Defaults within 1 year	90
Defaults within 2 years	150
Defaults within 3 years	200
FinBERT signal coverage	47,051 (74.2%)
LM signal coverage	47,051 (74.2%)

Panel B: Variable distributions

Variable	N	Mean	SD	P25	Median	P75
Total assets ($M)	63,449	6,354	27,909	119	555	2,609
Book leverage	63,449	0.234	0.236	0.018	0.185	0.366
Tobin's Q	60,926	1.788	1.750	0.779	1.214	2.090
Profitability	63,245	-0.010	0.315	-0.030	0.086	0.146
Cash/assets	63,448	0	0	0	0	0
FinBERT signal	47,051	-0.039	0.980	-0.651	-0.059	0.563
LM signal (PC1)	47,051	-0.003	0.991	-0.613	-0.025	0.583

Notes: Sample of US public firms (excluding financials and utilities) covering fiscal years 2006–2022. Default events combine LoPucki Bankruptcy Research Database filings (Chapter 7 and Chapter 11) with CRSP delisting codes 400–499 (liquidations) and 552 (insufficient capital). Defaults within h years are forward-looking from the observation date. Both text signals (FinBERT and LM) are standardized to mean zero and unit variance within each fiscal year. Financial variables are winsorized at the 1st and 99th percentiles within year.

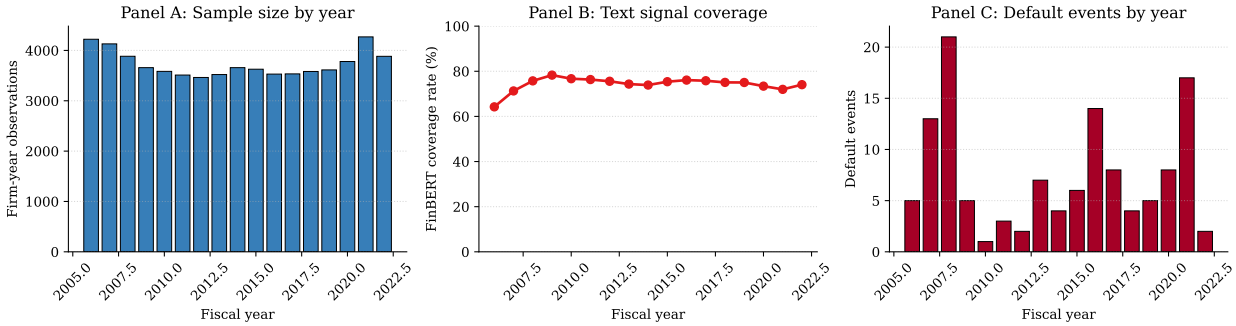


Figure 1: Sample composition over time. Panel A: number of firm-years per fiscal year. Panel B: FinBERT signal coverage rate per year (approximately 75% of firm-years have valid signals throughout the sample; coverage drops in early years as 10-K filings became electronically available, and in 2022 as the most recent filings have not yet been processed). Panel C: number of default events per year, showing concentration around the 2008–2010 financial crisis and the 2020 COVID shock.

panel; combined with 44 LoPucki bankruptcies (with no overlap), this yields 125 distress events, an annual incidence of 0.20%.

This empirical default rate substantially understates the true incidence of corporate default for two reasons. First, distressed firms frequently exit Compustat reporting before bankruptcy filing, and our requirement of complete financial data therefore induces survivor bias. Second, the LoPucki database covers only large bankruptcies (assets exceeding $100 million), missing smaller-firm default events. For purposes of structural estimation (Section 8), we therefore set the target default rate to 2% per annum, the historical average for US corporate defaults reported in Moody's Annual Default Study, rather than to the panel-observed rate.

3.3 Financial moments

For each firm-year we construct standard variables: book leverage D_{it}/A_{it} where D is total debt and A is total assets; investment rate $\Delta K_{it}/K_{i,t-1}$; cash ratio cash_{it}/A_{it}; profitability (operating income before depreciation over assets); and Tobin's Q (market value of equity plus debt over book assets). To address outliers, we winsorize all ratios at the 1st and 99th percentiles. Market capitalization is constructed from CRSP price and shares outstanding at fiscal-year close.

4 NLP Sentiment Analysis

For each firm-year (i, t) in the sample, I contruct text signals using two separate methods: s_{it}^{LM} from the Loughran-McDonald dictionary and s_{it}^{FB} from a FinBERT projection. Both are standardized to mean zero and standard deviation one. The pipeline for extracting and converting public 10-K disclosures into a text signal proceeds as follows.

4.1 Filing identification

For each unique CIK in the sample, I query SEC EDGAR via the `edgartools` Python package (Gunning, 2024) for all 10-K filings between 2006–2023, retrieving accession numbers, filing dates, and form types.

4.2 Text extraction

For each identified filing, we extract Item 1A (Risk Factors) and Item 7 (Management's Discussion & Analysis). Extraction proceeds in two stages: a primary parser using `edgartools`' built-in section parser, accepting parsed sections of length ≥ 500 characters; and a fallback regex-based parser on the raw HTML when the primary parser fails. We locate Item 1A as "Item 1A" followed by "Risk Factors" and terminate at "Item 1B" or "Item 2"; Item 7 boundaries are defined analogously. Once the text is extracted, it is then cleaned by removing HTML entities, dropping table-like rows, and normalizing white space.

4.3 LM dictionary scoring

The Loughran-McDonald (LM) dictionaries are curated word lists specifically designed for financial text. They were created by Tim Loughran and Bill McDonald, who demonstrated that general-purpose sentiment dictionaries — like the Harvard General Inquirer that psychologists and political scientists use — badly misclassify financial language. Loughran and McDonald went through financial text manually and built word lists that discriminate between positive and negative financial contexts. In the LM dictionary, words are classified into six sentiment categories:

- **Negative**: words indicating unfavorable conditions ("default", "bankruptcy", "litigation", "delinquent", etc.)
- **Positive**: words indicating favorable conditions ("achieve", "attain", "efficient", "profitable")
- **Uncertainty**: words indicating vagueness or lack of predictability ("approximate", "contingent", "unpredictable", "volatile")
- **Litigious**: words associated with legal proceedings and regulatory risk ("arbitration", "deposition", "injunction", "lawsuit")
- **Strong Modal**: words expressing strong conviction or obligation ("never", "shall", "will", "certainly")
- **Weak Modal**: words expressing hedging or equivocation ("apparently", "could", "might", "possibly", "probably")

For each filing we tokenize Item 1A and Item 7, I then count the occurrences words from Loughran-McDonald dictionary. More specifically, for each category c we compute the fraction

$$f_{it}^{c} = \frac{\sum\{\text{words in category } c\}}{\sum\{\text{total words}\}} \tag{1}$$

applied separately to Item 1A, Item 7, and the combined text.

4.4 FinBERT

FinBERT is a modified version of Google's BERT (Bidirectional Encoder Representations from Transformers). BERT gives each word a different vector depending on the surrounding sentence; so "bank" in a credit risk paragraph gets a different embedding than "bank" in a geography paragraph. BERT works by reading text in both directions simultaneously, reading words to the left and right of each position to understand context. When BERT is passed through a sentence, it outputs a 768-dimensional vector for each token in the sentence, plus a single 768-dimensional vector representing the whole sentence. The issue with BERT is that it is primarily trained on general English texts, where words differ systematically in sentiment and meaning to their financial equivalents. As such, FinBERT is a BERT model that is specifically trained on financial text, which allows FinBERT

to distinguish between financial sentences like "the firm faces material risk of covenant violation" (negative, credit-relevant) and "the firm reported material improvements in operating efficiency" (positive, performance-relevant) in a way that the original BERT model struggles to do. For this paper, we specifically use a verision of FinBERT from `ProsusAI/finbert`. Upon doing so, I cap each filing at 500 sentences for computational tractability. The document embedding $e_{it} \in \mathbb{R}^{768}$ is then the mean of these sentence embeddings.

4.5 Signal construction

To construct the LM-based signal, I use the first principal component of three category fractions in the combined text: *negative*, *uncertainty*, and *weak modal*. To remove time trends in disclosure style (10-K filings have grown longer and more boilerplate over time), each category is first standardized within each fiscal year:

$$\tilde{f}_{it}^{c} = \frac{f_{it}^{c} - \bar{f}_{t}^{c}}{\sigma_{t}^{c}} \tag{2}$$

I then perform PCA on these three standardized fractions and take the first principal component, standardized again within year. By construction, the final text signal s_{it}^{LM} has mean zero and unit variance within each year; higher values correspond to greater use of negative, uncertain, or hedging language.

For Finbert, the 768-dimensional document embedding cannot enter our scalar-signal model directly. To reduce this down to a one-dimensional text signal value, I project this embedding onto a "default-risk direction" identified through supervised learning. To avoid look-ahead bias, where our analysis uses information on defaults that wouldn't have been available at the point in time being analysed, we split the sample temporally. Observations from the first half of the sample's time range are used as a training set, and the projection is applied to all firm-years (training and test). On the training sub-sample I estimate a logistic regression of the form

$$\Pr(\mathrm{default}_{i,t+1} = 1 \mid e_{it}) = \sigma\big(\beta^{\top} e_{it} + \alpha\big) \tag{3}$$

Given the density and correlation of FinBERT embeddings, we include an L2 regularization penalty ($C = 1.0$), along with a inverse-class frequency weighting. The fitted coefficient vector $\hat{\beta} \in \mathbb{R}^{768}$ defines the default-risk direction. The FinBERT signal is the standardized projection s_{it}^{FB}.
it should be further noted that these LM and FinBERT signals are not constructed identically: LM is unsupervised, being PCA of dictionary fractions; whereas FinBERT is supervised, being a learned projection on financial distress event data toward a default-risk direction. We address this asymmetry directly in Section 5.1 by constructing a supervised LM variant when performing reduced-form analysis.

5 Reduced-Form Evidence

I test whether the constructed text signals predict subsequent corporate default. The dependent variable is $D_{i,t+h} \in \{0, 1\}$, an indicator equal to one if firm i experiences a default event within h years of fiscal year t. Our default specification uses $h = 2$ years; we present robustness across horizons in Section **??**. We estimate linear probability models of the form

$$D_{i,t+h} = \alpha + \gamma s_{it} + X_{it}^{\top}\delta + \mu_j + \tau_t + \varepsilon_{it} \tag{4}$$

where $s_{it} \in \{s_{it}^{\mathrm{LM}}, s_{it}^{\mathrm{FB}}\}$ is the text signal of interest, X_{it} is a vector of standard financial controls (log assets, book leverage, Tobin's Q, profitability, and cash ratio), μ_j is a 1-digit SIC industry fixed effect, and τ_t is a year fixed effect. Standard errors are clustered at the firm level.

Table 2 presents the FinBERT regressions. We observe that the signal s^{FB} is significant at the 1% level. while controls and FEs reduces the point estimate from +0.0048 to +0.0037, a roughly 23% reduction, the coefficient remains economically meaningful and statistically robust.

Table 2: Baseline FinBERT regressions.

	(1)	(2)	(3)	(4)	(5)
s^{FB}	$+0.0048^{***}$	$+0.0048^{***}$	$+0.0042^{***}$	$+0.0036^{***}$	$+0.0037^{***}$
	(0.0007)	(0.0007)	(0.0007)	(0.0006)	(0.0006)
Log assets		$+0.0003^{*}$	+0.0000	−0.0002	−0.0002
Book leverage			$+0.0088^{***}$	$+0.0092^{***}$	$+0.0090^{***}$
Tobin's Q			-0.0005^{***}	-0.0005^{***}	-0.0005^{***}
Profitability			−0.0003	+0.0006	+0.0008
Cash/assets			+0.0023	$+0.0030^{*}$	$+0.0034^{*}$
Industry FE	No	No	No	Yes	Yes
Year FE	No	No	No	No	Yes
Observations	41,154	41,154	40,715	40,715	40,715
R-squared	0.0085	0.0086	0.0100	0.0177	0.0187

Notes: The dependent variable is default within two years (binary). The FinBERT signal s^{FB} is the projection of [CLS] embeddings of 10-K text onto the default direction, standardized to mean zero and unit variance within each fiscal year. Coefficients give the marginal effect on the probability of default within two years per standard deviation increase in s^{FB}. Industry FE use 1-digit SIC. Cluster-robust standard errors in parentheses, clustered by firm (gvkey). Stars indicate statistical significance: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

This estimate implies a one-standard-deviation increase in the FinBERT signal is associated with a 0.37 percentage point increase in 2-year default probability after partialling out standard financial controls and absorbing all common variation across industries and years. Against an unconditional 2-year default rate of 0.24% in our sample, this is a very large effect. Then, on table 3 I present the analogous regressions for the LM scores. The contrast with Table 2 is stark: the LM coefficient is small (−0.0001 to −0.0006), wrong-signed in some specifications, and not

statistically significant. The unsupervised dictionary signal does not predict default.

Table 3: Baseline LM regressions.

	(1)	(2)	(3)	(4)	(5)
s^{LM}	−0.0001	+0.0001	+0.0003	+0.0004	+0.0004
	(0.0003)	(0.0003)	(0.0004)	(0.0004)	(0.0004)
Log assets		+0.0004***	+0.0001	−0.0002	−0.0002
Book leverage			+0.0123***	+0.0122***	+0.0122***
Tobin's Q			−0.0009***	−0.0008***	−0.0008***
Profitability			−0.0027*	−0.0007	−0.0005
Cash/assets			+0.0005	+0.0016	+0.0019
Industry FE	No	No	No	Yes	Yes
Year FE	No	No	No	No	Yes
Observations	41,154	41,154	40,715	40,715	40,715
R-squared	0.0000	0.0003	0.0042	0.0141	0.0150

Notes: The dependent variable is default within two years (binary). The LM signal s^{LM} is the first principal component of the standardized fractions of negative, uncertainty, and weak-modal Loughran-McDonald (2011) categories in the combined Item 1A (Risk Factors) and Item 7 (MD&A) sections of the 10-K. Both signals and PC weights are computed within year. Coefficients give the marginal effect on the probability of default within two years per standard deviation increase in s^{LM}. Industry FE use 1-digit SIC. Cluster-robust standard errors in parentheses, clustered by firm (gvkey). Stars indicate statistical significance: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Across multiple horizons, Table 4 repeats the full-controls + FE specification at one-, two-, and three-year horizons. The FinBERT coefficient grows with the horizon, consistent with text containing medium-term information that gradually materializes in default events. Similarly to before, the LM coefficient remains statistically insignifcant from zero across all horizons.

To assess economic significance non-parametrically, we sort firm-years into quintiles by within-year FinBERT signal and report the default rate within each quintile. Table 5 and Figure 2 report the result, where we find that the default rate in the top FinBERT quintile (most negative-toned 10-Ks) is 1.12%: roughly 50 times the default rate in the bottom quintile (0.024%). The top quintile contains 86% of all default events in this subsample, despite being only 20% of firm-years. This implies that the default risk is heavily concentrated in the most-extreme firm-years.

A natural concern is that FinBERT and LM signals may carry redundant information, with our test simply detecting one of two essentially equivalent measures. Table 6 addresses this by including both signals in the same regression. The FinBERT coefficient is essentially unchanged when the LM signal is added. The LM coefficient is small and noisy throughout.

Another concern is that FinBERT may be a poor substitute for a market-based distress measure such as Merton-naive distance-to-default. We construct a Merton-naive distance-to-default following Bharath & Shumway (2008) to test whether FinBERT remains informative once we control for it. Results are in Table 7, where Merton distress predicts default on its own (+0.0018, $p < 0.01$). Once financial controls and fixed effects are added, however, Merton distress becomes

Table 4: FinBERT predictive power across forecast horizons.

	1-year	2-year	3-year
s^{FB}	$+0.0023^{***}$	$+0.0037^{***}$	$+0.0048^{***}$
	(0.0004)	(0.0007)	(0.0009)
s^{LM}	-0.0001	-0.0002	-0.0003
	(0.0002)	(0.0004)	(0.0005)
Log assets	-0.0002	-0.0002	-0.0000
Book leverage	$+0.0053^{***}$	$+0.0089^{***}$	$+0.0115^{***}$
Tobin's Q	-0.0002^{***}	-0.0005^{***}	-0.0006^{***}
Profitability	$+0.0004$	$+0.0008$	$+0.0009$
Cash/assets	$+0.0024^{*}$	$+0.0035^{**}$	$+0.0043^{**}$
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	40,715	40,715	40,715
R-squared	0.0161	0.0187	0.0214

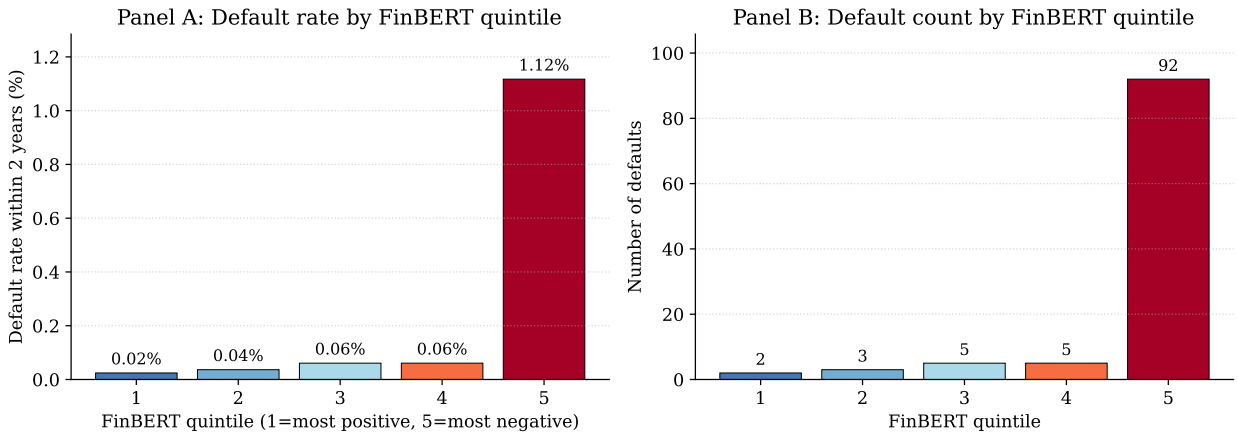
Notes: The dependent variable is default within h years where $h \in \{1, 2, 3\}$ across columns. All specifications include both s^{FB} and s^{LM}, financial controls, industry FE (1-digit SIC), and year FE. Coefficients give the marginal effect on default probability per standard deviation increase in the signal. Cluster-robust standard errors in parentheses, clustered by firm (gvkey). Stars indicate statistical significance: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: Default concentration by FinBERT quintile.

FinBERT quintile	Mean signal	N firm-years	Defaults within 2 years	Default rate (%)	Share of defaults (%)
1 (lowest)	-1.392	8,238	2	0.024	1.9
2	-0.533	8,226	3	0.036	2.8
3	-0.062	8,229	5	0.061	4.7
4	+0.422	8,226	5	0.061	4.7
5 (highest)	+1.314	8,235	92	1.117	86.0
Total	-0.050	41,154	107	0.260	100.0

Notes: Sample restricted to firm-years with both FinBERT signal and observable forward default. Quintiles are assigned within each fiscal year so that sample composition does not drive the result. The top quintile (5) captures the firm-years with the most negative-toned 10-K text. The default rate within two years rises monotonically across quintiles, with the top quintile concentrating a disproportionate share of subsequent default events.

Figure 2: Default concentration in the top FinBERT quintile.



Notes: Panel A: default rate (within 2 years) by FinBERT quintile. Panel B: count of default events by FinBERT quintile. Quintile assignment is within year.

Table 6: Horse race between FinBERT and LM text signals.

	(1)	(2)	(3)	(4)	(5)
s^{LM}	-0.0001		-0.0006^{*}	-0.0004	-0.0002
	(0.0003)		(0.0004)	(0.0004)	(0.0004)
s^{FB}		$+0.0048^{***}$	$+0.0049^{***}$	$+0.0043^{***}$	$+0.0037^{***}$
		(0.0007)	(0.0007)	(0.0007)	(0.0007)
Log assets				$+0.0000$	-0.0002
Book leverage				$+0.0086^{***}$	$+0.0089^{***}$
Tobin's Q				-0.0005^{***}	-0.0005^{***}
Profitability				-0.0004	$+0.0008$
Cash/assets				$+0.0026$	$+0.0035^{**}$
Industry FE	No	No	No	No	Yes
Year FE	No	No	No	No	Yes
Observations	41,154	41,154	41,154	40,715	40,715
R-squared	0.0000	0.0085	0.0086	0.0100	0.0187

Notes: The dependent variable is default within two years (binary). Both signals are standardized to mean zero and unit variance within each year; coefficients are thus per-standard-deviation marginal effects on default probability. Industry FE use 1-digit SIC. Cluster-robust standard errors in parentheses, clustered by firm (gvkey). Stars indicate statistical significance: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

statistically insignificant (-0.0006, ns), while FinBERT remains strongly significant throughout ($+0.0035$, $p < 0.01$ in column 5). This implies that our text signal carries information about default that is not subsumed by a classical market-based distress measure.

Table 7: Incremental predictive power of FinBERT over Merton-naive distance-to-default (Bharath and Shumway, 2008). The dependent variable is default within two years (binary). Merton distress is defined as $-DD$, where $DD = [\ln(V/F) + (r - 0.5\sigma_V^2)T]/(\sigma_V\sqrt{T})$ with V equity plus book debt, F default point $= dlc + 0.5 \cdot dltt$, $T = 1$ year, r the 1-year Treasury yield, and σ_V the Bharath-Shumway naive approximation. Both Merton distress and s^{FB} are standardized to mean zero and unit variance within each year. Coefficients are marginal effects on default probability per standard deviation increase. Industry FE use 1-digit SIC.

	(1)	(2)	(3)	(4)	(5)
Merton distress ($-DD$)	$+0.0018^{***}$		$+0.0010^{**}$	-0.0005	-0.0006
	(0.0004)		(0.0005)	(0.0006)	(0.0006)
s^{FB}		$+0.0048^{***}$	$+0.0045^{***}$	$+0.0042^{***}$	$+0.0035^{***}$
		(0.0007)	(0.0008)	(0.0007)	(0.0007)
Log assets				$+0.0002$	$+0.0000$
Book leverage				$+0.0120^{***}$	$+0.0118^{***}$
Tobin's Q				-0.0006^{***}	-0.0006^{***}
Profitability				-0.0036^{*}	-0.0025
Cash/assets				-0.0012	$+0.0000$
Industry FE	No	No	No	No	Yes
Year FE	No	No	No	No	Yes
Observations	43,689	41,154	34,018	33,996	33,996
R-squared	0.0014	0.0085	0.0087	0.0112	0.0153

Notes: Cluster-robust standard errors in parentheses, clustered by firm (gvkey). Stars indicate statistical significance: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

One remaining issue is over fitting and the signals power for out-of-sample prediction. That is, it's possible that the FinBERT signal might predict in-sample because the signal itself was constructed via supervised learning on default labels. While I use a temporal split when constructing the signal, the in-sample test does not rule out information leakage from the construction procedure into the regression test. To address this, I perform a chronological out-of-sample evaluation: fit logit models on fyear ≤ 2014, predict on fyear ≥ 2015, and compare test-set the area under the Receiver Operating Characteristics (AUC) curve (a plot that shows the trade-off between capturing true defaulting firms and falsely flagging safe firms) across model specifications.

I present results in Table 8, where we see that the test-set AUC rises from 0.80 (controls only) to 0.92 (controls + FinBERT); a substantial gain that holds in genuinely out-of-sample data. The LM signal helps modestly (AUC 0.85), but the bulk of the predictive gain comes from FinBERT, and adding LM to FinBERT gives essentially no further improvement. This rules out the "signal construction is just overfitting" interpretation: FinBERT's predictive power persists out-of-sample.

Table 8: Out-of-sample prediction performance.

	Controls (1)	$+s^{\mathrm{LM}}$ (2)	$+s^{\mathrm{FB}}$ (3)	+ both (4)
Test-set AUC	0.7984	0.8476	0.9166	0.9174
Test-set avg. precision	0.0279	0.0705	0.0360	0.0354
In-sample AUC (train)	0.8668	0.9025	0.9386	0.9385
N (train)	30,056	23,023	23,023	23,023
N (test)	22,610	17,692	17,692	17,692
Defaults (test)	61	54	54	54

Notes: Logistic regression models fit on firm-years with fyear ≤ 2014 and evaluated on firm-years with fyear ≥ 2015. Controls are log assets, book leverage, Tobin's Q, profitability, and cash/assets. The text signals s^{LM} and s^{FB} are standardized within year. The dependent variable is default within two years. AUC = area under the ROC curve; AP = average precision (area under precision-recall curve).

5.1 Supervised LM

The LM-vs-FinBERT comparison so far has been between an unsupervised LM signal (PC1 of dictionary fractions) and a supervised FinBERT signal (logistic projection of embeddings onto default direction). One criticism of this analysis is that the gap is driven by the supervision, not by the difference between dictionary and neural representations. To resolve this concern, I construct a supervised LM signal using the same projection procedure as FinBERT: a logistic regression of next-period default on the three LM category fractions, with an L2 regulation penalty and balanced class weights, trained on fyear ≤ 2013 (the median fiscal year of the text-signal sample). The resulting projection is then applied to all firm-years and standardized within year.

On Table 9, I present this comparison. The supervised LM coefficient (-0.0001) is essentially identical to the unsupervised version (-0.0001); both are not significantly different from zero. By contrast, FinBERT remains strongly positive and significant ($+0.0048$, $p < 0.01$). When all three signals are included with controls and fixed effects (column 4), the supervised LM coefficient is -0.0003 and not significant; FinBERT is $+0.0037$ and $p < 0.01$. The gap between FinBERT and LM is not closed by supervising LM.

5.2 Identification considerations

The coefficient γ above only has a causal interpretation conditional on the assumption that text language is exogenous, conditional on the controls. This exogeneity does not necessearily hold, as the relationship between language and default may reflect unobserved firm characteristics that correlate with both. Our reduced-form analysis therefore documents an informational link between text and subsequent default rather than a structural channel. The structural model (Section 6) provides one rationalization of this link by treating the signal as informative about latent productivity which becomes part of a competitive lenders information set.

Table 9: Fair comparison of LM dictionary and FinBERT signals.

	(1)	(2)	(3)	(4)
$s^{\text{LM,unsup}}$ (PCA)	-0.0001			
	(0.0003)			
$s^{\text{LM,sup}}$		-0.0001		-0.0003
		(0.0003)		(0.0004)
s^{FB}			$+0.0048^{***}$	$+0.0037^{***}$
			(0.0007)	(0.0007)
Log assets				-0.0002
Book leverage				$+0.0089^{***}$
Tobin's Q				-0.0005^{***}
Profitability				$+0.0008$
Cash/assets				$+0.0036^{**}$
Industry FE	No	No	No	Yes
Year FE	No	No	No	Yes
Observations	41,154	41,154	41,154	40,715
R-squared	0.0000	0.0000	0.0085	0.0188

Notes: The supervised LM signal ($s^{\text{LM,sup}}$) is constructed by projecting the three LM category fractions (negative, uncertainty, weak modal) onto the default direction identified by logistic regression on firms in the first half of the sample (fyear $\leq$ median), using the same procedure as s^{FB}. The unsupervised variant $s^{\text{LM,unsup}}$ is the first principal component of the same three categories, standardized within year. All signals are standardized to mean zero and unit variance within each year. The dependent variable is default within two years (binary). Industry FE use 1-digit SIC.Cluster-robust standard errors in parentheses, clustered by firm (gvkey). Stars indicate statistical significance: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6 Augmented Hennessy-Whited Model

6.1 Setup

Time is discrete and infinite. The firm chooses period-t capital k_{t+1} and one-period bond issuance b_{t+1} to maximize the expected discounted value of dividends to equity holders. The firm's state at the beginning of period t is the triple $(\tilde{w}_t, z_t, s_t)$:

- $\tilde{w}_t$ is the firm's net worth (cash plus assets less debt obligations) at the start of the period, after carrying over last period's realized profit and any accumulated savings.
- z_t is current productivity, evolving as a log-AR(1): $\ln z_{t+1} = \rho \ln z_t + \varepsilon_{t+1}$ with $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.
- s_t is a public text signal observed at the start of t, correlated with z_{t+1} via the relationship

$$s_t = \lambda_{\text{text}} \ln z_{t+1} + (1 - \lambda_{\text{text}})\eta_t \tag{5}$$

  where $\eta_t \sim \mathcal{N}(0, \sigma_\eta^2)$ and $\lambda_{\text{text}} \in [0, 1]$ governs informativeness.

The augmented model nests HW07 as the special case $\lambda_{\text{text}} = 0$: then s_t is pure noise and provides no incremental information over z_t, which leads us back to the original model specification from HW07.

6.2 Production and capital

Output in period t is Cobb-Douglas: $y_t = z_t k_t^\alpha$ where $\alpha \in (0, 1)$ is the returns-to-scale parameter, calibrated as 0.627 in HW07. Capital depreciates at rate $\delta = 0.15$ per period, while gross investment is $i_t = k_{t+1} - (1 - \delta)k_t$.

6.3 Debt and bond pricing

The firm issues one-period debt b_{t+1} at the beginning of period t, repayable at the start of period $t + 1$ at the gross interest rate $1 + r_t^{\text{bond}}$. The competitive lender prices the bond to break even in expectation:

$$b_{t+1} = (1 + r_t^{\text{bond}})^{-1} \cdot \mathbb{E}_t\big[b_{t+1} \cdot \mathbb{1}_{\text{no default}} + R_{t+1} \cdot \mathbb{1}_{\text{default}}\big] \tag{6}$$

where the expectation is taken using the lender's posterior over z_{t+1}, conditional on (z_t, s_t), and R_{t+1} is the recovery in default. The text signal enters through this conditional expectation: a more pessimistic s_t shifts the lender's posterior toward lower z_{t+1}, raises default probability.

6.4 Realized net worth and default

At the start of period $t + 1$, productivity z_{t+1} realizes. The firm's net worth becomes

$$\tilde{w}_{t+1} = (1 - \delta)k_{t+1} + z_{t+1}k_{t+1}^\alpha - T_c\big(z_{t+1}k_{t+1}^\alpha - \delta k_{t+1} - r_t^{\text{bond}} b_{t+1}\big) - (1 + r_t^{\text{bond}})b_{t+1} \tag{7}$$

where T_c is asymmetric corporate tax (rates $\tau_c^+ = 0.40$ on positive income, $\tau_c^- = 0.20$ on negative income, the latter reflecting limited loss carry-back).

The firm defaults if $\tilde{w}_{t+1} < \underline{w}(z_{t+1})$, which is an endogenously-determined default threshold. The threshold depends only on productivity, not on net worth, since at the boundary the firm is indifferent between continuing (with value $V(\underline{w}, z, \cdot) = 0$) and defaulting. Once a firm defaults, equity holders receive zero and lenders recover

$$R_{t+1} = (1-\xi)(1-\delta)k_{t+1} + z_{t+1}k_{t+1}^{\alpha} - T_c\big(z_{t+1}k_{t+1}^{\alpha} - \delta k_{t+1}\big) - \underline{w}(z_{t+1} \tag{8}$$

where ξ is the bankruptcy cost, estimated in HW07 at 0.104.

6.5 Equity issuance and payouts

After paying for capital and debt at the start of t, the firm has financing need $k_{t+1} - \tilde{w}_t - b_{t+1}$. If positive, the firm issues equity at cost $\Lambda(x) = \lambda_1 x + \lambda_2 x^2$ for $x > 0$ (the fixed cost λ_0 from HW07 is dropped to ensure smoothness of the period payoff for gradient-based DEQN training). If negative, the firm distributes a dividend, taxed at the asymptotic rate $\bar{\tau}_d$ via the convex distribution-tax function $T_d(X) = \bar{\tau}_d X - (\bar{\tau}_d/\phi)(1 - \exp(-\phi X))$ where ϕ is HW07's tax-convexity parameter.

6.6 Bellman equation

The firm's value function satisfies

$$V(\tilde{w}_t, z_t, s_t) = \max_{k_{t+1} \geq 0,\, b_{t+1}} \Big\{ \pi(\tilde{w}_t, k_{t+1}, b_{t+1}) + \beta \mathbb{E}_t\big[V(\tilde{w}_{t+1}, z_{t+1}, s_{t+1})\big] \Big\} \tag{9}$$

where $\pi(\cdot)$ is the period payoff (distribution net of tax less issuance net of cost), $\beta = 1/(1+r(1-\tau_i))$ is the after-personal-tax discount factor, and the expectation is over both z_{t+1} (which determines $\tilde{w}_{t+1}$ via the realized-net-worth equation) and s_{t+1} (which is conditionally normal given z_{t+1}). The boundary condition is $V(\underline{w}(z), z, s) = 0$ for all (z, s), capturing the equity-holder's indifference between continuation and default at the boundary.

6.7 Parameters

We adopt HW07 Table I full-sample estimates as our baseline parametrization. The augmented parameters λ_{text} and σ_η are the focus of structural estimation (Section 8). The full parameter vector is: $(\alpha, \rho, \sigma_\varepsilon, \lambda_1, \lambda_2, \xi, \phi, \lambda_{\text{text}}, \sigma_\eta) = (0.627, 0.684, 0.118, 0.091, 0.0004, 0.104, 0.732, \cdot, \cdot)$, with λ_{text} initialized at 0.3 and σ_η normalized to 1.0.

7 DEQN Solution Method

7.1 Approach

The state space $(\tilde{w}, z, s)$ is three-dimensional and the Bellman equation involves a non trivial expectation over two-dimensional next-period shocks (z', s'), plus an optimal default decision at each state. While this model is originally solved over a grid search, standard discretization becomes computationally heavy at our state dimension, especially because the value function has a kink at the default boundary. Instead, following Maliar et al. (2021) and Azinovic et al. (2022), I use a deep equilibrium network (DEQN) approach, where I parameterize value, policy, and default-threshold functions as neural networks and minimize the Bellman residual (plus boundary and monotonicity penalties) by gradient descent.

7.2 Network architecture

We use three separate networks:

- `ValueNet`: $V(\tilde{w}, z, s) \to \mathbb{R}$, a 3-input MLP with two hidden layers of 16 units and tanh activation.
- `PolicyNet`: $(\tilde{w}, z, s) \to (k', b')$, a 3-input MLP with output dimension 2. The k' output passes through softplus to ensure positivity.
- `DefaultNet`: $z \to \underline{w}(z)$, a 1-input MLP with output passed through softplus to ensure negativity, since the boundary $\underline{w}$ is structurally negative.

States are normalized to approximately unit variance before being fed to the networks via a state normalizer that stores sample means and standard deviations.

7.3 Training

The training objective combines four loss terms:

1. **Bellman residual**: squared deviation between $V(\tilde{w}, z, s)$ predicted by the network and the right-hand side of the Bellman equation, with the next-period expectation estimated by Monte Carlo.

2. **Boundary loss**: penalizes $V(\underline{w}(z), z, s) \neq 0$.

3. **Monotonicity penalty**: penalizes $\partial V / \partial z < 0$ (i.e, higher productivity must weakly raise value).

4. **Payout anchor**: at high net worth ($\tilde{w} > 30$), penalizes positive equity issuance, addressing a tendency of the solver to converge to "hoarding" equilibria where firms issue new equity even when fully self-financed.

We pre-train all three networks on simple analytical targets for 500 epochs to prevent the joint training from diverging in the early iterations. After this, I then jointly minimize the combined loss for 1500–2500 epochs using Adam (a specific optimization algorithm for DEQN solvers) with a learning rate of 10^{-3}.

7.4 Smooth approximations

The default decision is an indicator, while the realized net worth at the boundary a max. To enable gradient-based training, we replace these with smooth approximations: $\mathbb{1}_{\text{default}} \approx \sigma(s(\underline{w} - \tilde{w}_{t+1}))$ with sharpness $s = 5$, and $\max(\underline{w}, \tilde{w}_{t+1}) \approx \text{logsumexp}$ with the same sharpness. Smoothing parameters are then verified to be tight enough that the trained policy approximately satisfies the discrete default rule.

7.5 Properties of the solution

The trained network produces an equilibrium with the following qualitative features at HW07 baseline parameters: default frequency 5–10% across the state space; default threshold $\underline{w}(z)$ in the range $[-9, -7]$ for $z \in [0.5, 2]$; positive borrowing at low net worth (firms with $\tilde{w} = -10$ choose $b' \approx +11$); savings at high net worth ($b' < 0$ for $\tilde{w} > 20$); and positive distributions at high $\tilde{w}$. These match HW07's qualitative predictions.

7.6 Limitations of the DEQN approach

There are two known limitations to be addressed. Firstly, our trained equilibrium features higher cash holdings than typical HW07 estimates, which is likely an unresolved calibration issue rather than a model failure. Secondly, the DEQN solver does not provide global convergence guarantees; we have verified the solution is locally stable but cannot rule out alternative equilibria.

8 SMM Estimation

8.1 Approach

We attempt to estimate the augmented model by simulated method of moments (SMM), where for each candidate parameter vector θ, the model is solved with DEQN, a panel of firms is simulated, and then compute 22 moments from the simulated panel, along with minimizing the weighted distance to data moments. The optimization is done by Nelder-Mead over $\theta \in \mathbb{R}^9$.

8.2 Moments

Our 22 moments comprise three groups:

- **Financial moments**: cross-sectional means, standard deviations, and AR(1) coefficients of book leverage, investment rate, cash ratio, and profitability; plus the unconditional default rate target.
- **Pricing moments**: means of bond spreads relative to the risk-free rate at five rating buckets (BBB, BB, B, CCC), aimed at identifying the credit-risk parameters.
- **Text moments**: correlations of the text signal with current productivity (s_t vs. $\ln z_t$), next-period default (s_t vs. D_{t+1} for both FinBERT and LM), and conditional default rates in top vs. bottom signal quintiles. These are the moments designed to identify the text-informativeness parameter λ_{text}.

Weights are diagonal: $W_{ii} = 1/m^2_{\text{data},i}$, putting each moment on a comparable percent-deviation scale, with a floor at 10^{-4} for moments near zero.

8.3 Default rate target

The data default rate is 0.20% per annum (panel-observed), which substantially understates the true incidence due to survivor bias (Section 3). For the SMM target we use Moody's historical 2% per annum, documenting this as a calibration choice.

8.4 Initialization and bounds

Parameters are initialized at HW07 Table I full-sample estimates, with $\lambda_{\text{text}} = 0.3$ and $\sigma_\eta = 1.0$. The bounds are as follows: $\alpha \in (0.3, 0.9)$, $\rho \in (0.3, 0.95)$, $\sigma_\varepsilon \in (0.05, 0.30)$, $\lambda_1 \in (0.01, 0.30)$, $\lambda_2 \in (0.0001, 0.01)$, $\xi \in (0.05, 0.30)$, $\phi \in (0.1, 0.9)$, $\lambda_{\text{text}} \in (0, 1)$, $\sigma_\eta \in (0.3, 2.0)$.

8.5 Estimation results

A single SMM run using Nelder-Mead with 500-firm $\times$ 100-period simulated panels reduced loss from 228.5 at the initial point to 40.8 over 236 function evaluations before terminating on maximum iterations. Loss decrease was concentrated in around the first 80 iterations, with marginal improvements from iteration 85 onward, indicating that the simplex collapsed in a region where the objective is effectively flat.

Only one parameter moves meaningfully: the capital share α drops from 0.627 to 0.434. All other parameters, including the the added parameter of interest λ_{text}, remain within 10% of their initial values. The λ_{text} estimate of 0.303 is essentially identical to its starting value of 0.300, moving by less than 1% over 236 function evaluations.

This pattern is suggestive of weak identification, as the optimizer cannot find a direction in which reducing loss requires moving λ_{text} away from its starting value. As such, This result is tested using the following two additional diagnostics.

Table 10: SMM parameter estimates (9 parameters, 22 moments)

Parameter	HW07 initial	SMM estimate	Movement
α	0.627	0.434	−0.193
ρ	0.684	0.699	+0.015
σ_ε	0.118	0.125	+0.007
λ_1	0.091	0.100	+0.009
λ_2	0.0004	0.0004	+0.0000
ξ	0.104	0.109	+0.005
ϕ	0.732	0.729	−0.003
λ_{text}	**0.300**	**0.303**	**+0.003**
σ_η	1.000	1.067	+0.067

8.6 Multi-start diagnostic

To determine if the lack of movement in λ_{text} is a genuine finding or the result of starting at $\lambda_{\text{text}} = 0.3$, we re-ran the full SMM with alternative starting values ($\lambda_{\text{text}} \in \{0.1, 0.5\}$) while keeping all other initial parameters unchanged. In both cases, λ_{text} remains near its initial value through the iterations that the optimizer meaningfully explores, consistent with the hypothesis that the loss surface is flat in the λ_{text} direction rather than having a minimum near 0.3.

8.7 One-dimensional line search

For a sharper diagnostic, we hold the non-text parameters fixed at their SMM estimates and evaluate loss over an 11-point grid of $\lambda_{\text{text}} \in \{0.0, 0.1, \ldots, 1.0\}$, with the random seed held fixed across grid points to isolate deterministic variation from Monte Carlo noise. We perform this line search at two simulation sizes: a baseline of $N = 500$ firms $\times$ $T = 100$ periods, and a larger specification with $N = 2{,}000 \times T = 200$ (8$\times$ more observations).

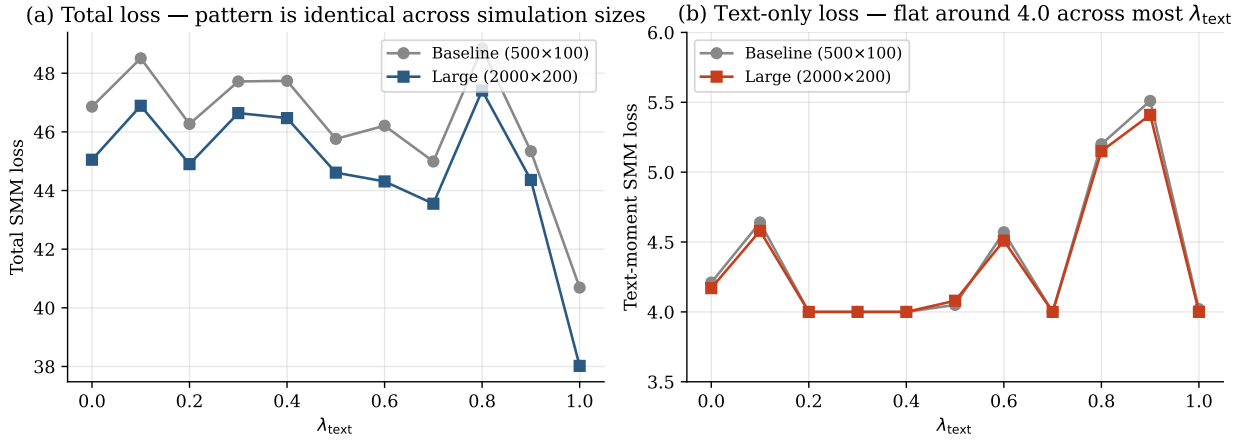
Results are reported in Table 11 and Figure 3.

Table 11: Line-search loss across λ_{text} at two simulation sizes

	Baseline (500×100)		Large (2000×200)	
λ_{text}	Total loss	Text-only loss	Total loss	Text-only loss
0.0	46.86	4.21	45.05	4.17
0.1	48.51	4.64	46.89	4.58
0.2	46.27	4.00	44.90	4.00
0.3	47.72	4.00	46.64	4.00
0.4	47.74	4.00	46.47	4.00
0.5	45.76	4.06	44.61	4.08
0.6	46.21	4.57	44.31	4.51
0.7	44.99	4.00	43.55	4.00
0.8	48.85	5.20	47.41	5.15
0.9	45.34	5.51	44.36	5.41
1.0	40.69	4.02	38.02	4.00

Our first observation is that the text-only loss is flat at 4.00 across most of the grid ($\lambda_{\text{text}} \in$

Figure 3: Line search over λ_{text} at two simulation sizes



Notes: Left: total SMM loss. Right: text-moments-only loss.

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$\{0.2, 0.3, 0.4, 0.7, 1.0\}$ all yield identical text-only loss), with three localized bumps at $\lambda_{\text{text}} \in \{0.1, 0.8, 0.9\}$. The pattern is neither monotonic nor single-peaked (U-shaped). Total loss similarly lacks monotonic or single-peaked structure.

Second, and more importantly, the 8× larger simulation produces a text-only loss curve that is *nearly identical* to the baseline (right panel of Figure 3). If MC noise in the signal-default correlation moments were the cause of the flat loss surface, we presume increasing the number of simulated observations by a factor of eight should reveal previously hidden signal. Because it does not reveal any signal, I conclude that the flat loss surface is deterministic – a feature of the moment-parameter mapping in this model at our parametrization — rather than a sampling product.

8.8 Interpretation

The three diagnostics (multi-start SMM, line search at baseline simulation size, line search at 8× simulation size) all point to the the parameter λ_{text} not being identified from the 22-moment specification we examine. I interpret this as a finding about moment design rather than about the economic meaning of text signals:

- The text-targeting moments in our specification (m17–m22) are correlations involving default, which is a rare event in our panel (125 events among 63,449 firm-years). Thus, the high variance may propagate into the simulated moments even at large simulation sizes.
- The structural link from λ_{text} to these moments is indirect: λ_{text} governs the posterior precision about z', which enters the bond price, which affects realized net worth, which determines default.
- Financial moments (leverage, investment, cash, Q) dominate the total loss. Changes in λ_{text}

can produce small effects on these moments via the solver's re-optimization, but these effects are swamped by the much larger financial-moment residuals.

Other possible candidates are not examined here but I flag for future work include: bond spreads conditional on text signal deciles; cross-sectional variance of signals within productivity-size bins; and moments based on text-implied disagreement across filings in the same industry-year cell. The primary empirical evidence for the text–default link in this thesis is therefore reduced-form (Section 5), with the structural framework providing an economic rationale rather than a sharp quantitative estimate of λ_{text}.

9 Limitations and Caveats

Across both our reduced-form and structural estimation, following are viewed as the main limitations of this methodology.

Survivor bias in default. As noted in Section 3, our panel-observed default rate (0.20%) substantially understates the true incidence of corporate default because financially distressed firms exit Compustat reporting before filing for bankruptcy. Ideally, I would have like to have used Moody's Default and Recovery Database, which contains Firm-level default events and recovery rates for calibration of default frequency and loss-given-default moments. However, this dataset is restricted-assess, resulting in us having to target the external Moody benchmark of 2%.

Conservative-firm equilibrium. The DEQN solver converges to an equilibrium in which firms hold cash buffers larger than typically observed in HW07-style estimations. The economic mechanisms (positive borrowing at low net worth, dividend distribution at high net worth, productivity-dependent default thresholds) are present but the calibration of cash holdings differs from HW07. This is a known limitation of our approach, and reflects difficulties in replicating HW07 outside of a clear replication package.

Identification of text-signal parameters. The diagnostics in Sections 8.5–8.8 show that λ_{text} is not identified from our 22-moment specification. The structural framework of Section 6 should therefore be understood as providing an economic rationalization of the reduced-form text–default link documented in Section 5, rather than as yielding a sharp quantitative estimate of the text-informativeness parameter.

Sample period. The 2006–2022 period spans two major financial crises (2008 and 2020). The text-default relationship may operate differently in calmer periods. Our robustness checks (Section 5) split the sample but may not fully address out-of-period generalization.

Causal interpretation. The reduced-form coefficient on the text signal is not a structural object: it conflates the information content of language with unobserved firm characteristics that correlate with both language style and default risk. Our structural model treats the signal as informative about productivity, which is one rationalization, but is not the only possible channel.

10 Conclusion

In this paper, I find that sentiment extracted from 10-K filings using a pre-trained neural language model (FinBERT) predicts subsequent corporate default on a panel of US public firms, while traditional dictionary-based sentiment (Loughran-McDonald) does not. The result is statistically robust ($p < 0.01$ with full controls and fixed effects), economically meaningful (top-quintile firms account for 86% of defaults), generalizes out-of-sample, and is incremental to a market-based distress measure (Merton-naive distance-to-default). Constructing a supervised LM signal that mimics FinBERT's projection procedure does not close the gap, ruling out the possibility of supervision-based explanation for FinBERT's advantage.

To supplement and identify this reduced-form analysis, I then develop a text-augmented version of the Hennessey & Whited (2007) dynamic corporate finance model, in which lenders update beliefs about next-period productivity using a public text signal.

I then attempt structural estimation of this augmented model by simulated method of moments and find that the text-informativeness parameter is not identified from a 22-moment specification standard in the financial economics literature. Three diagnostic procedures (multi-start SMM, line searches at two simulation scales) agree on this conclusion, providing robustness to our findings. I conclude this as a finding about moment design: the standard HW07 moments are insufficient to identify text-channel parameters, and future work should design moments specifically for the text channel. As such, the reduced-form evidence is the paper's primary empirical contribution. The structural framework provides an economic rationalization, but does not deliver a sharp quantitative estimate of how informative the text is.

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